Habitat fragmentation



Barro Colorado Island in the Panama Canal of Central America was formed in 1914, when tropical rain forest was flooded by damming, creating an isolated fragment of forest surrounded by water. Since 1946, the area has been studied in detail by biologists of the Smithsonian Institution and elsewhere to determine the effects of this habitat fragmentation: species diversity on the island has declined, and top predators are among the most vulnerable species. In the US, studies of habitat fragmentation and its effects on diversity in the Florida Keys led to Robert MacArthur and E.O. Wilson's seminal *Theory of Island Biogeography* (1967).

From such environments, planners have learned important lessons about how to conserve species in isolated patches of habitat—sometimes in the midst of cities—that are set aside as reserves. Barro Colorado, and places like it, have also provided vital opportunities for study, where ecologists can explore how changing species diversity affects the functioning of an ecosystem at every level.

See also: Mutualisms • Keystone species • The ecosystem • Organisms and their environment • Invasive species